

a4a stochastic slice biomass index

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1 Data

1.1 Trimmed time series (from 1985)

[...]

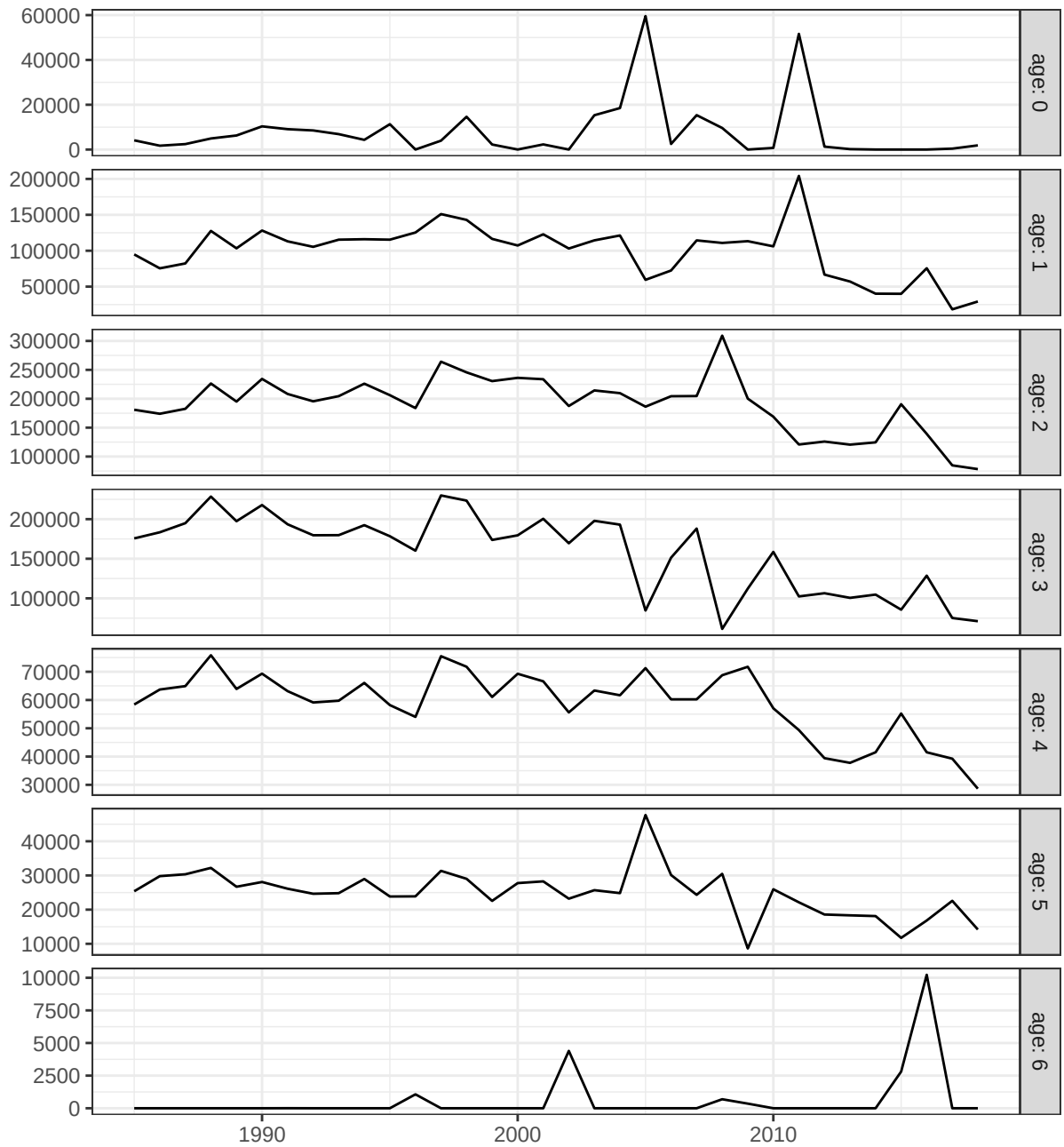


Figure 1. Time series of catch-at-age.

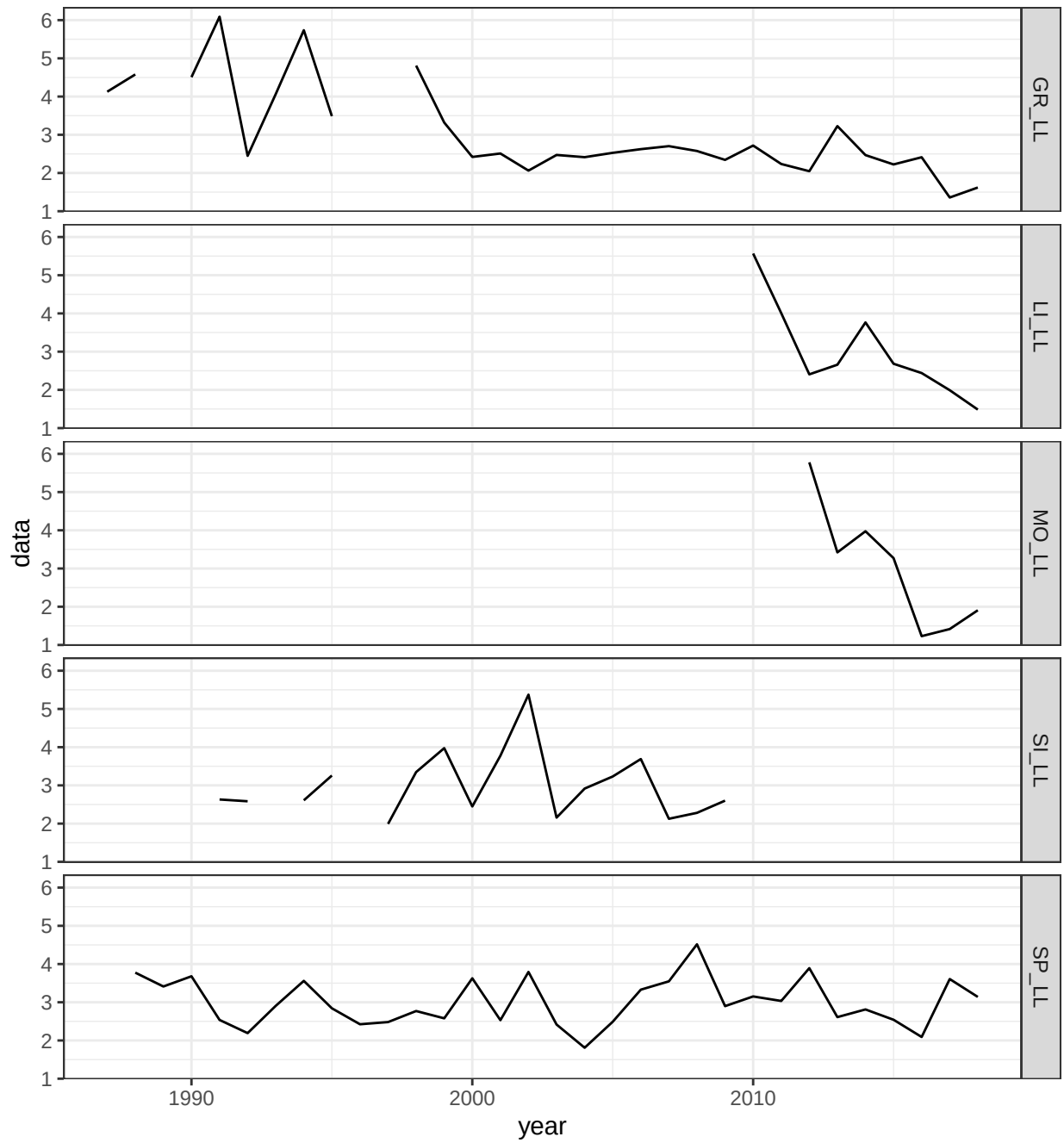
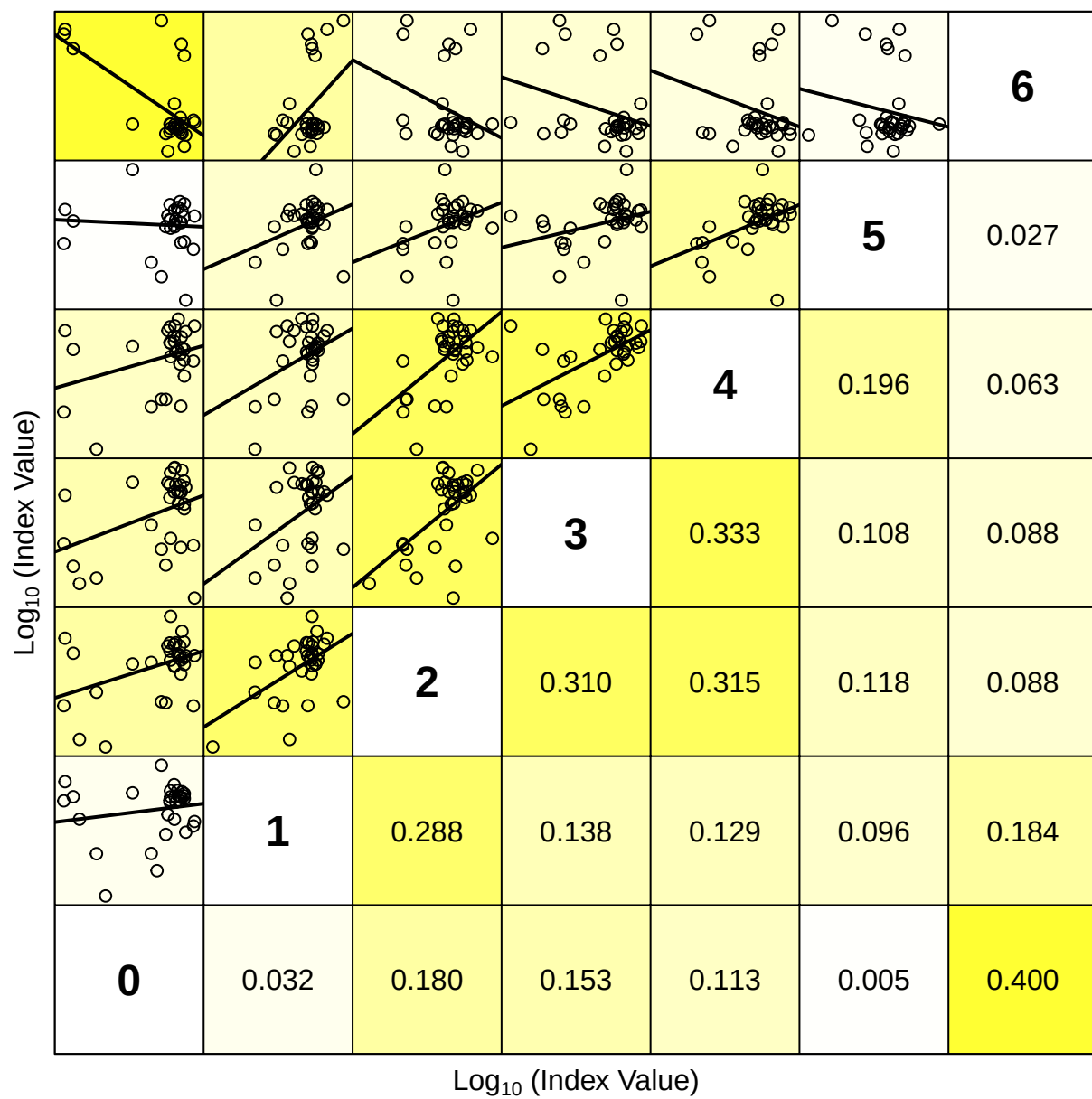


Figure 2. Biomass indices.



Lower right panels show the Coefficient of Determination (r^2)

2 Results fbar 2 - 4

Simple separable factor model on F results in a possible not positive definite covariance matrix which is an indicator of overfitting. The same stands for the built in tensor model. Catchability model is held constant through the years. Absence of SR model gave an unreasonable low recruitment (90.000). Some runs were tested by removing the spanish LLS, the results were almost identical.

2.1 Two “best” runs with a plus group of 6

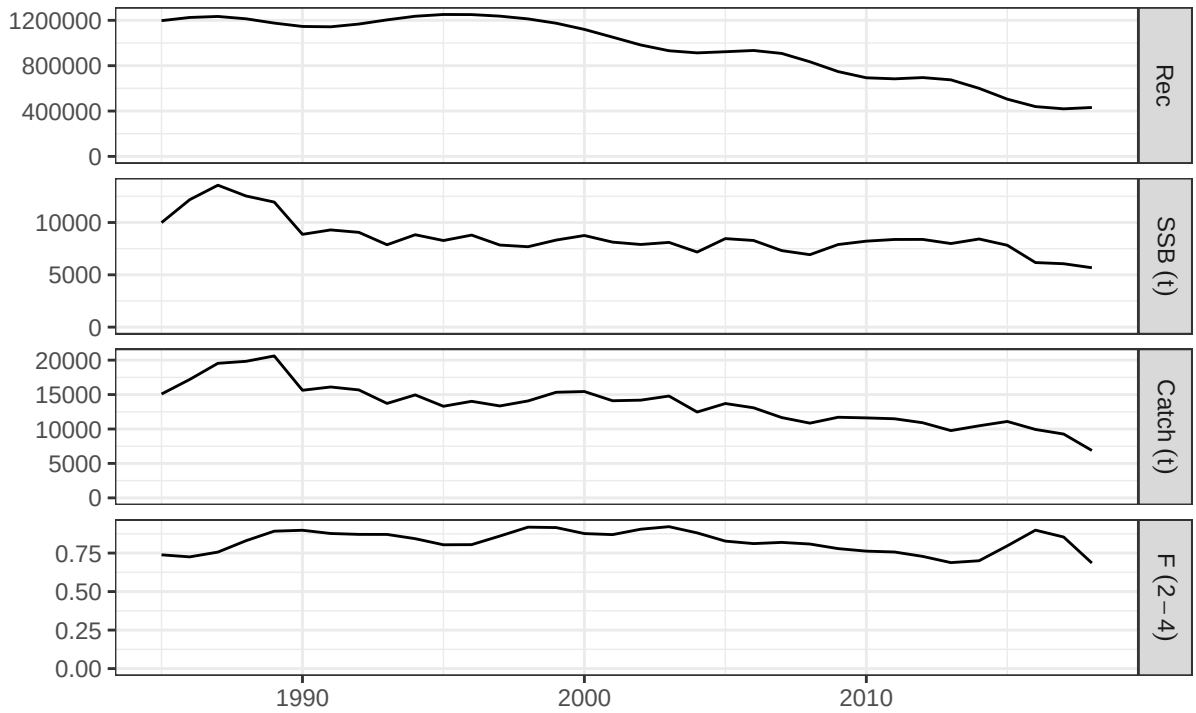


Figure 3. Results for fit 1. F model : $\sim, s(\text{year}, k = 17) + s(\text{age}, k = 3)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

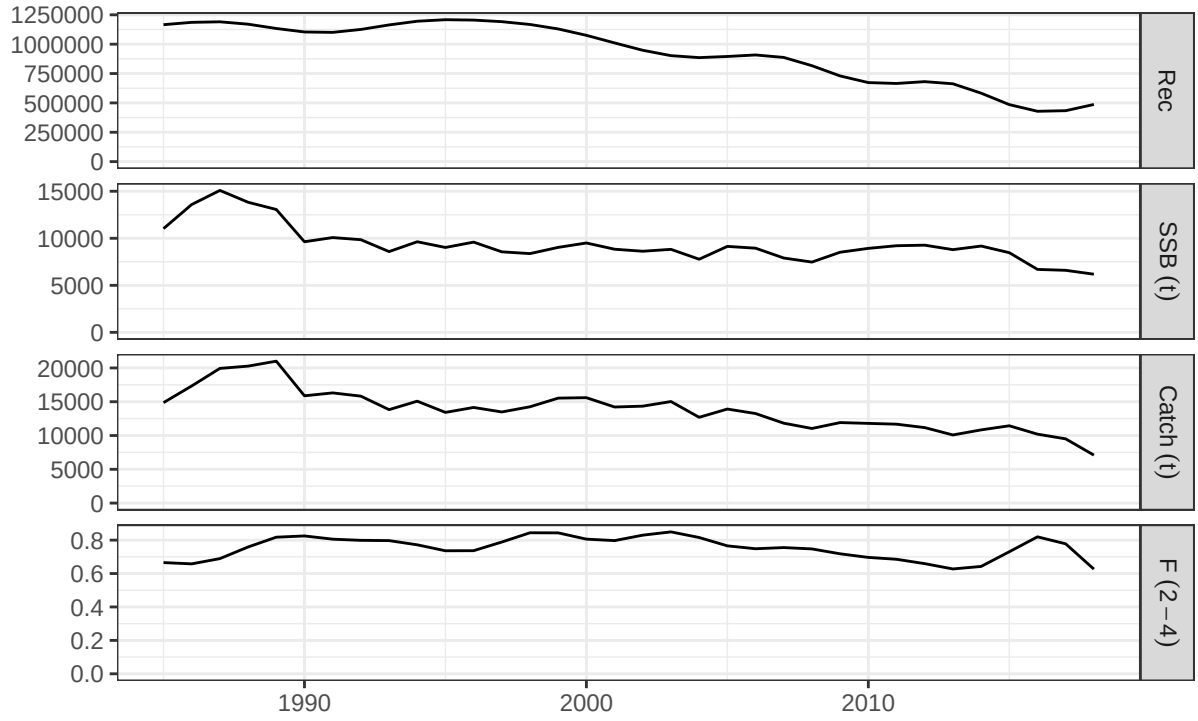


Figure 4. Results for fit 2. F model : $\sim, s(\text{year}, k = 17) + s(\text{replace}(\text{age}, \text{age} > 4, 4), k = 3)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

and the corresponding fishing mortalities:

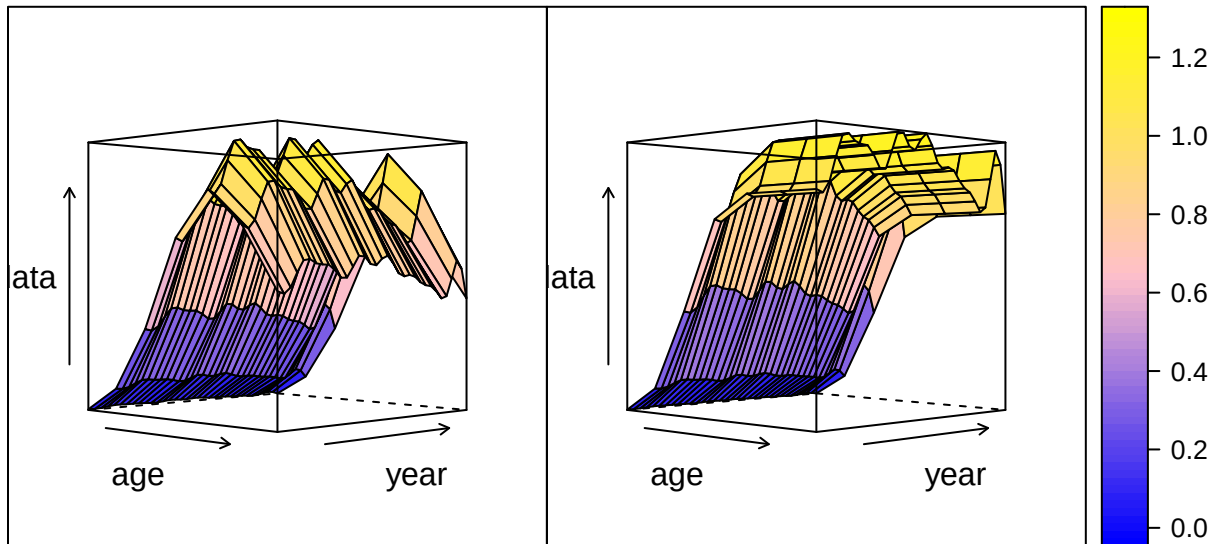


Figure 5. Harvest for fit1 (left) and fit2(right).

2.2 Diagnostics (bad fit in catch numbers in age 6)

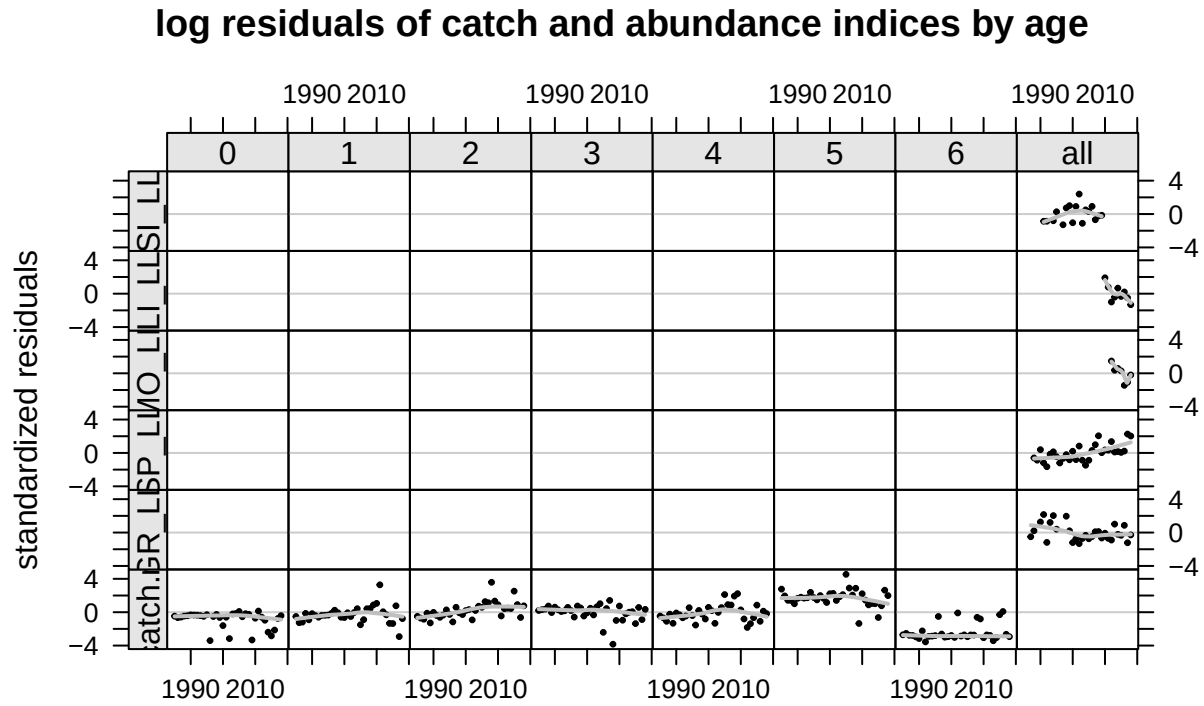


Figure 6. Residuals for fit 1.

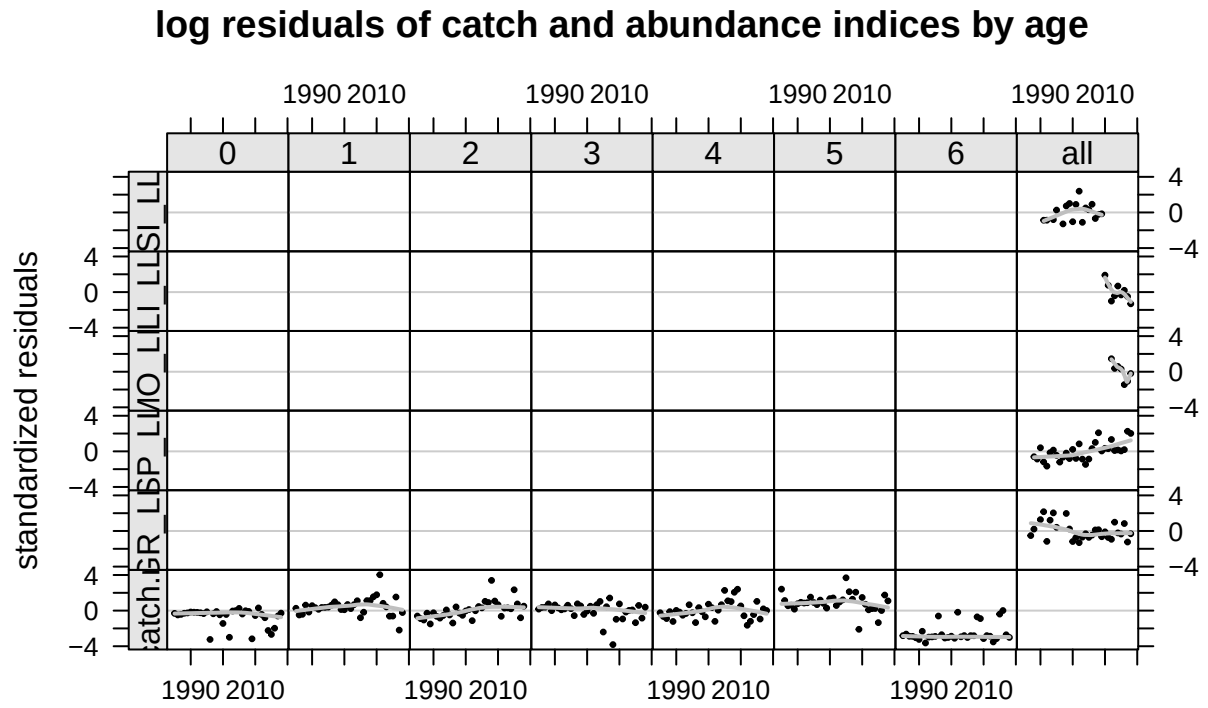


Figure 7. Residuals for fit 2.

2.3 Increase by one the knots on splines for age in fishing mortality model

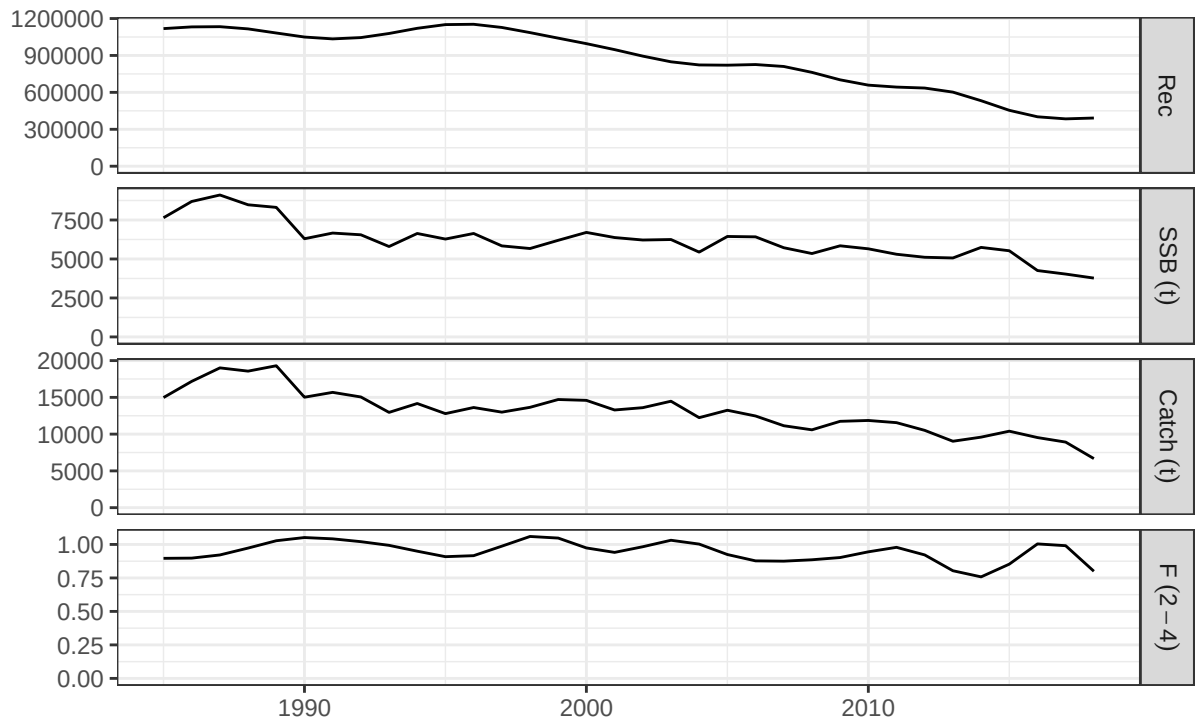


Figure 8. Results for fit 3. F model : $\sim, s(\text{year}, k = 17) + s(\text{age}, k = 4)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

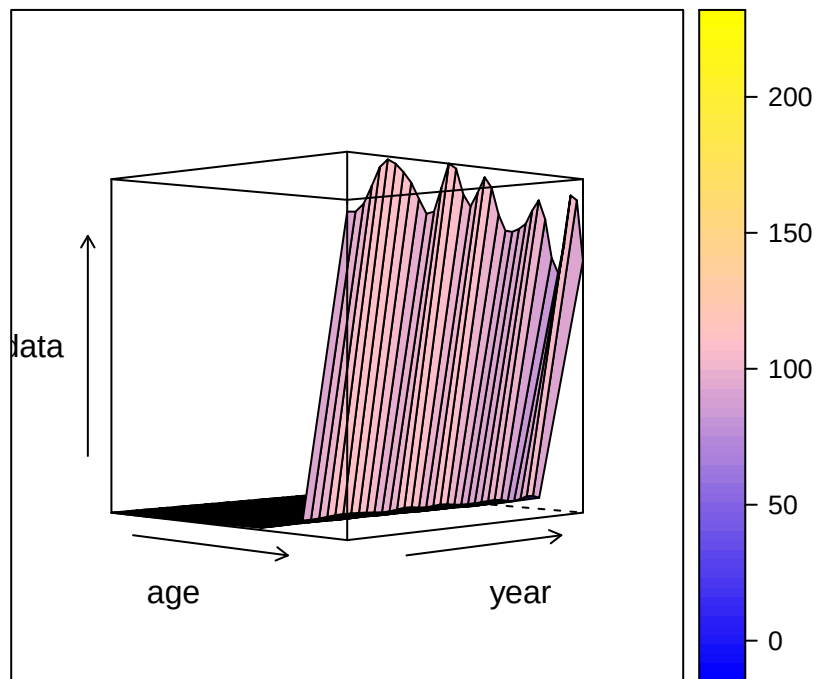


Figure 9. Fishing mortality for fit 3

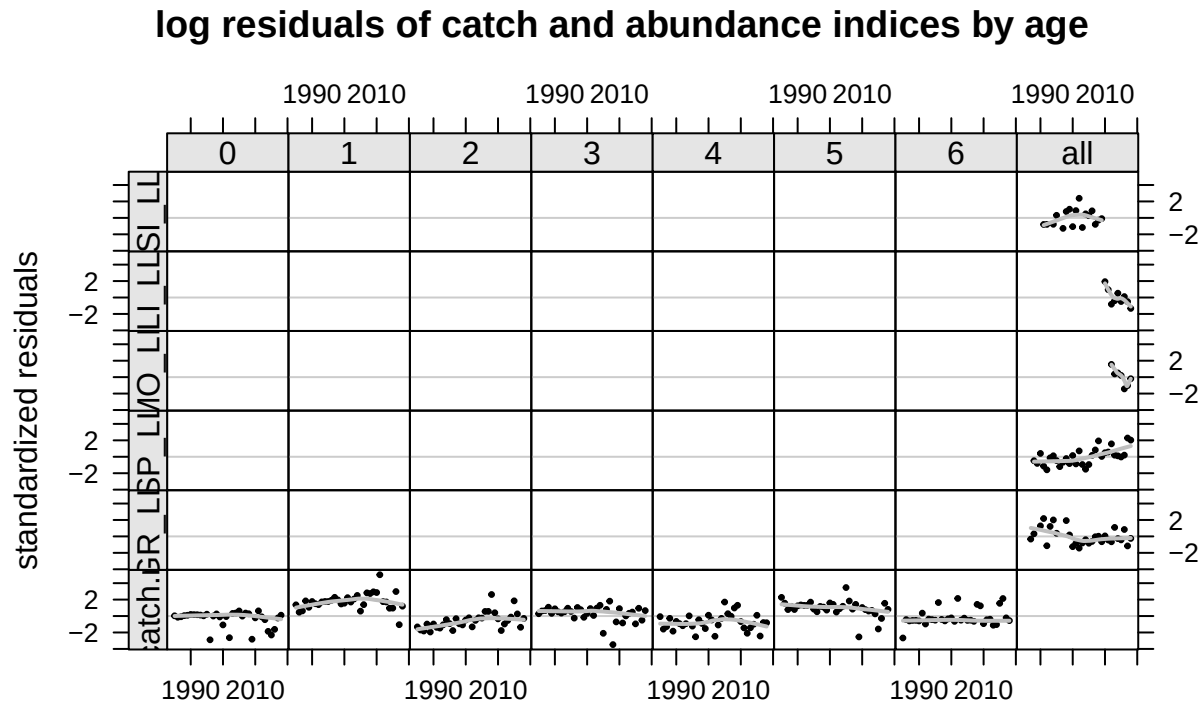


Figure 10. Residuals for fit 3.

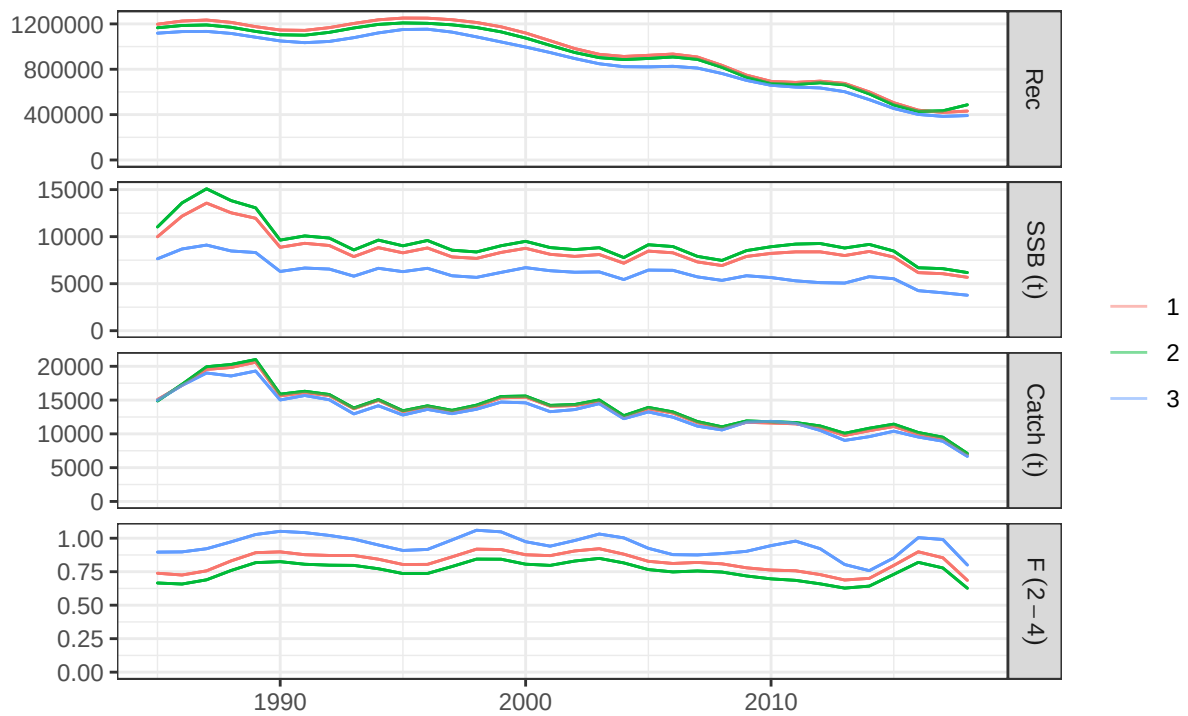


Figure 11. Results for the 3 fits.

2.4 Set plus group 5.

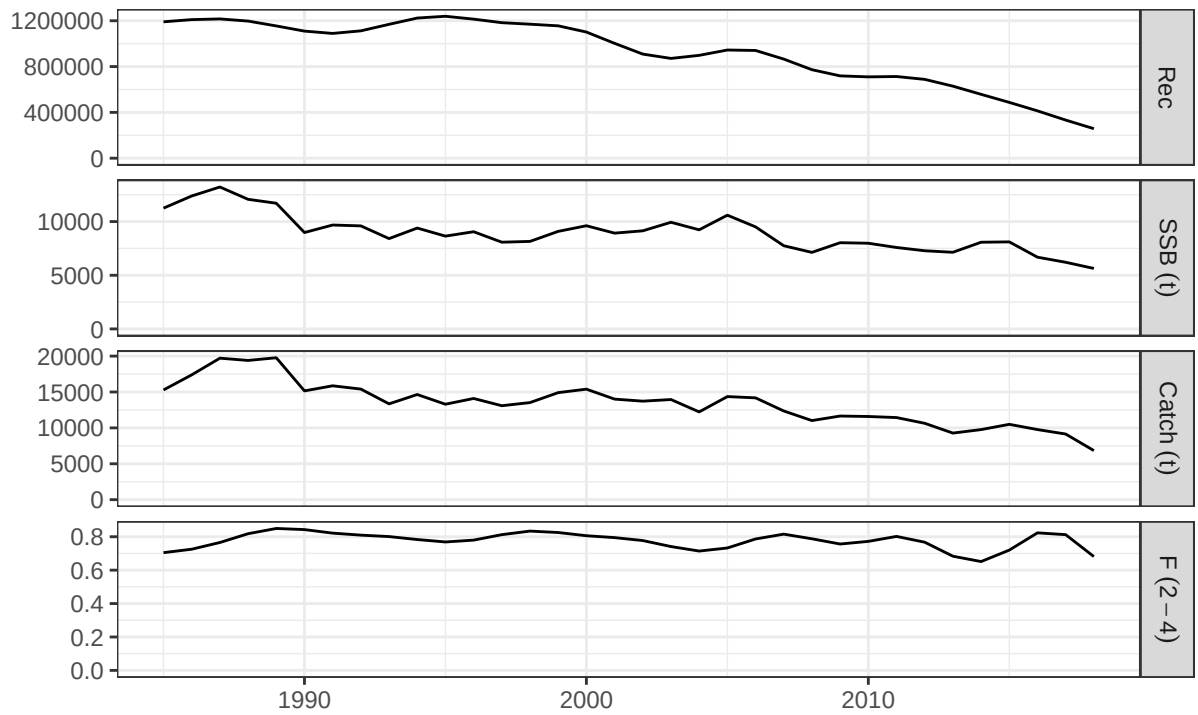


Figure 12. Results for fit 4. F model : $\sim, s(\text{year}, k = 17) + s(\text{age}, k = 3)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

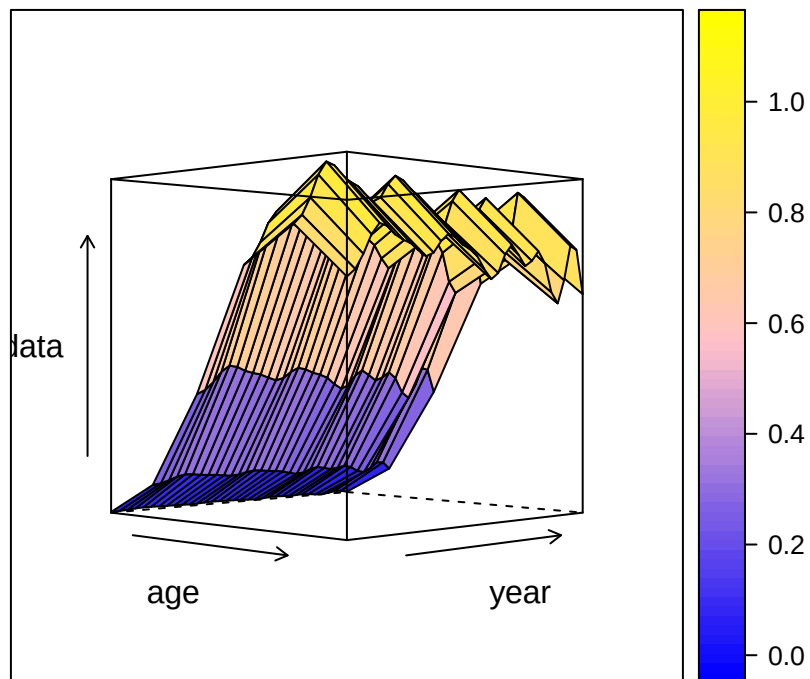


Figure 13. Fishing mortality for fit 4

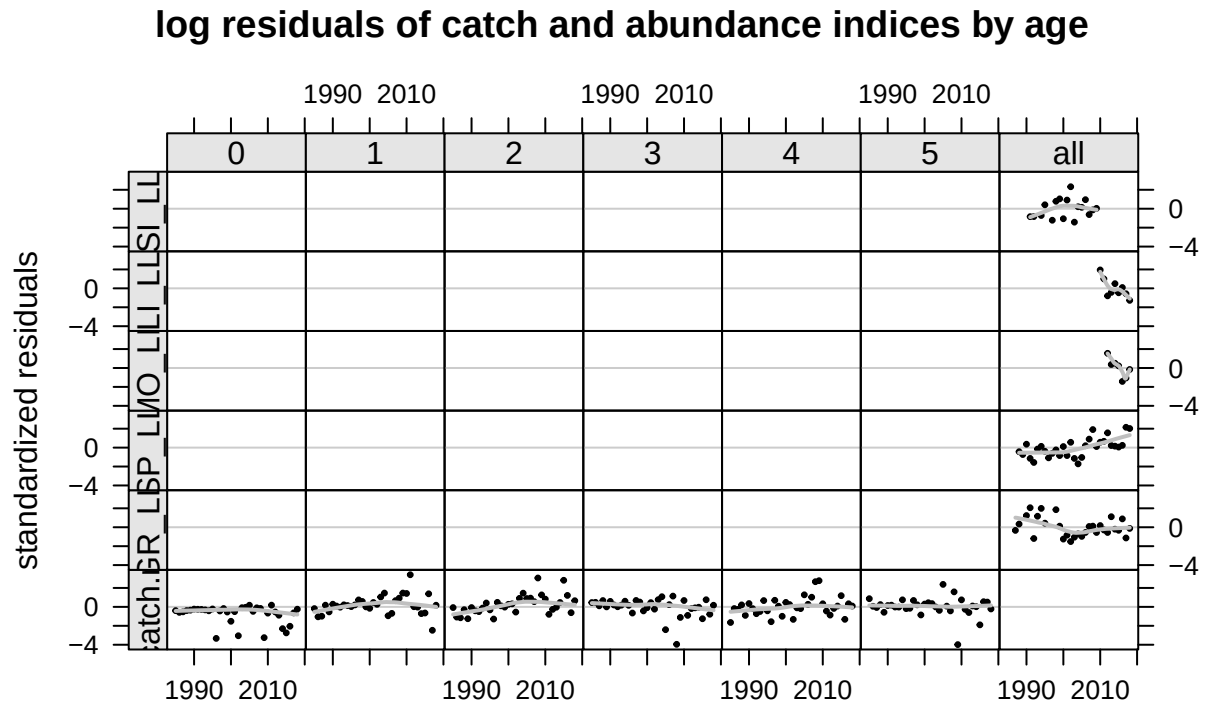


Figure 14. Residuals for 4.

2.5 same f after age 4

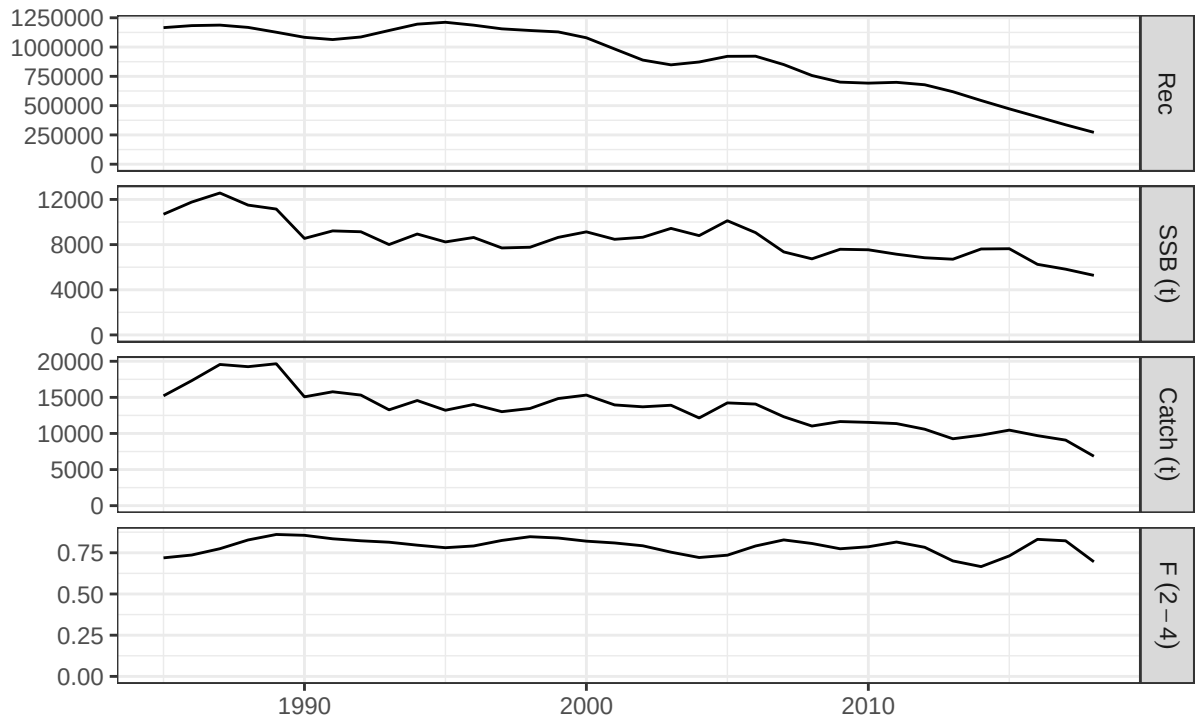


Figure 15. Results for fit 5. F model : $\sim, s(\text{year}, k = 17) + s(\text{replace}(\text{age}, \text{age} > 4, 4), k = 3)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

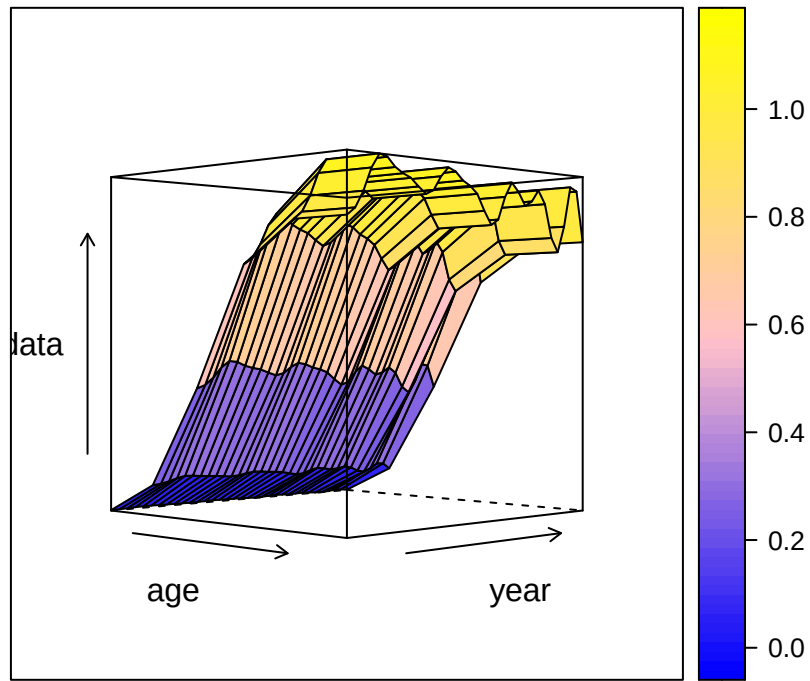


Figure 16. Fishing mortality for fit 5

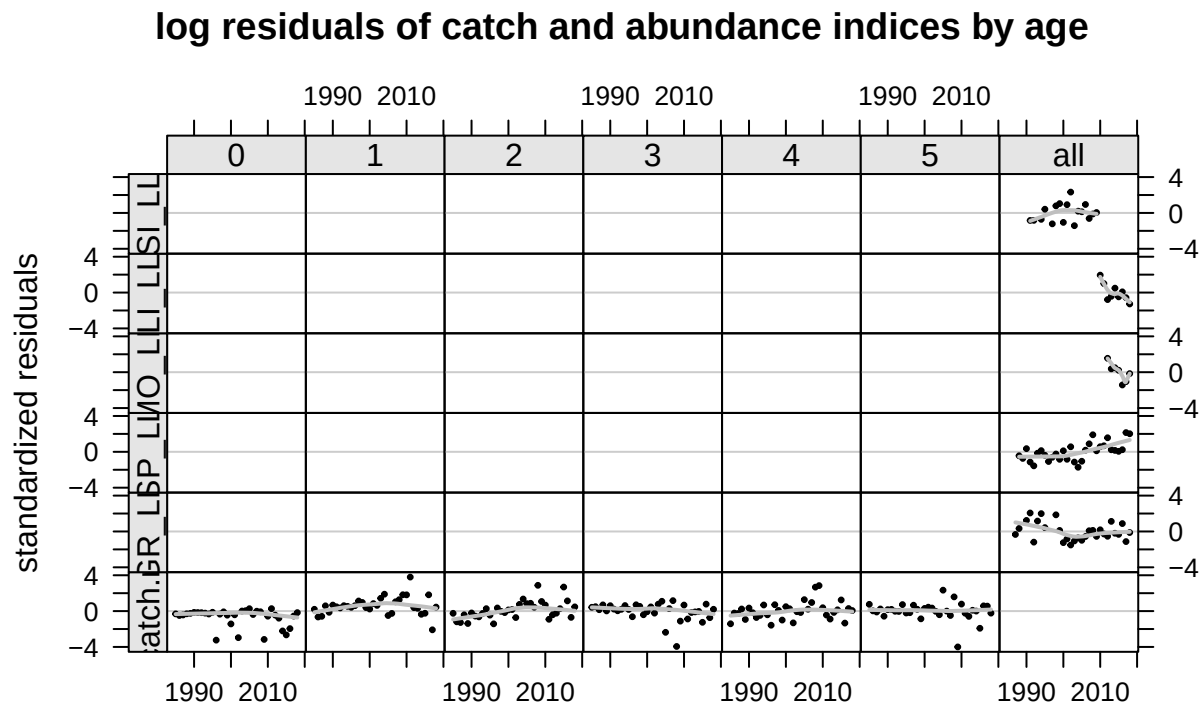


Figure 17. Residuals for 5.

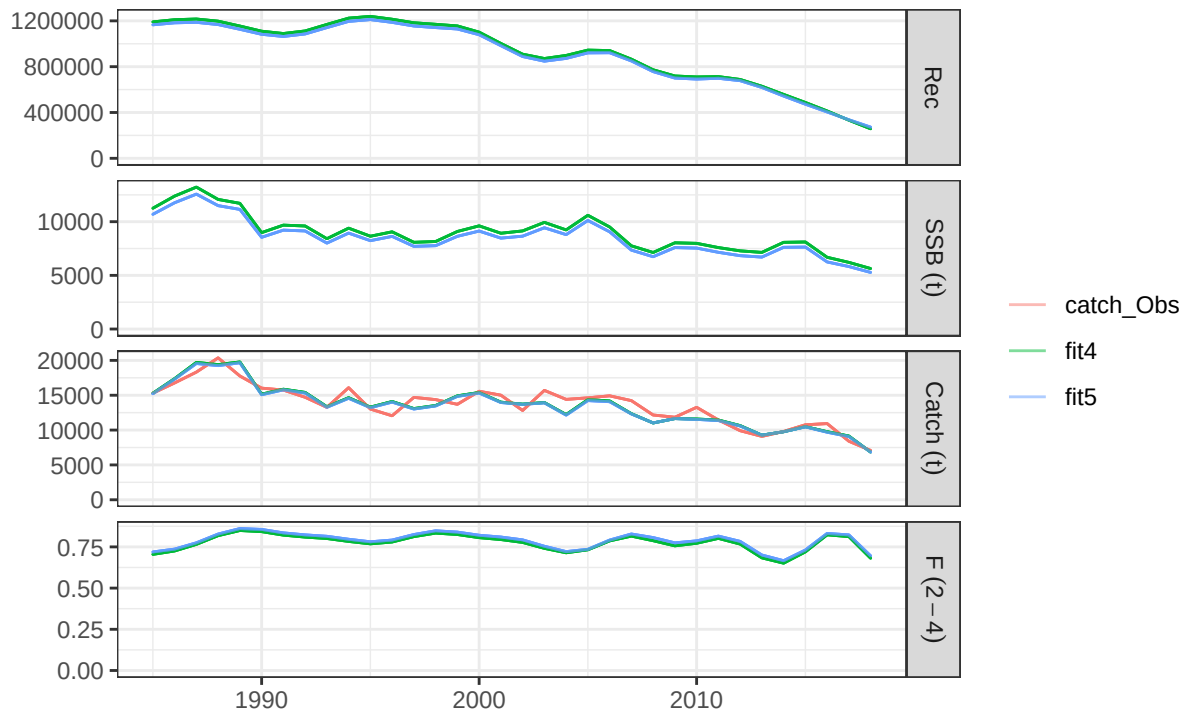


Figure 18. Results for the fits with plus group 5.

